

Homogeneous System of Linear Equations

Definition (1): The system of linear equations, $AX = 0$ is called a *homogenous system* and has

- Single *unique solution* (zero solution) if the matrix A is *nonsingular* ($|A| \neq 0$).
- Infinitely many solutions including the zero solution if A is *singular* ($|A| = 0$) or the number of variables is greater than the number of equations.

Dentition (2): A homogenous system *is consistent* because it has at least one solution,

Examples on Homogeneous System, $AX = 0$

Example (1): Solve the following system of equations.

$$x + 2y + 3z = 0$$

$$-x + 3y + 2z = 0$$

$$2x + y - 2z = 0$$

Solution

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ -1 & 3 & 2 & 0 \\ 2 & 1 & -2 & 0 \end{array} \right) \xrightarrow{R_1 + R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 5 & 5 & 0 \\ 2 & 1 & -2 & 0 \end{array} \right) \xrightarrow{-2R_1 + R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 5 & 5 & 0 \\ 0 & -3 & -8 & 0 \end{array} \right)$$

$$\xrightarrow{\frac{1}{5} R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -3 & -8 & 0 \end{array} \right) \xrightarrow{3R_2 + R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -5 & 0 \end{array} \right) \xrightarrow{-\frac{1}{5} R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

$$\xrightarrow{-R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \xrightarrow{-3R_3 + R_1} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \xrightarrow{-2R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right)$$

The system is consistent and has only the trivial solution (zero solution). i.e. $x = y = z = 0$

Note: The homogeneous System has only the zero solution if the matrix A is *nonsingular* ($|A| \neq 0$)

Example (2): Solve the following system of equations.

$$x + y - 3z = 0$$

$$2x + y - z = 0$$

$$3x + 2y - 4z = 0$$

Solution

$$\left(\begin{array}{ccc|c} 1 & 1 & -3 & 0 \\ 2 & 1 & -1 & 0 \\ 3 & 2 & -4 & 0 \end{array} \right) \xrightarrow{-2R_1 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & -3 & 0 \\ 0 & -1 & 5 & 0 \\ 3 & 2 & -4 & 0 \end{array} \right) \xrightarrow{-3R_1 + R_3} \left(\begin{array}{ccc|c} 1 & 1 & -3 & 0 \\ 0 & -1 & 5 & 0 \\ 0 & -1 & 5 & 0 \end{array} \right)$$

$$\xrightarrow{-R_2} \left(\begin{array}{ccc|c} 1 & 1 & -3 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & -1 & 5 & 0 \end{array} \right) \xrightarrow{R_2 + R_3} \left(\begin{array}{ccc|c} 1 & 1 & -3 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{-R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Since the last row contains zeros, therefore, the system **has an infinite number of solutions**.

Put $z = r$ (r is real number), then,

$$\text{From the 2-nd row, we get } y - 5z = 0 \Rightarrow y = 5z \Rightarrow y = 5r.$$

$$\text{From the 1-st row, we get } x + 2z = 0 \Rightarrow x = -2z \Rightarrow x = -2r.$$

Therefore, the solution set is : $(x = -2r, y = 5r, z = r)$.

Note: The homogeneous System, **has an infinite number of solutions**, if the matrix **A is singular matrix** ($|A| = 0$) or, the number of **variables (unknowns)** is greater than the number of equations.

Example (3): Solve the following system of equations.

$$x + 2y - z + w = 0$$

$$-x - 2y + 3z + 5w = 0$$

$$-x - 2y - z - 7w = 0$$

Solution

Here, in this example we notice that the number of variables (unknowns) is greater than the number of equations, then, the system is **consistent and has an infinite number of solutions**

Therefore,

$$\begin{aligned} & \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ -1 & -2 & 3 & 5 & 0 \\ -1 & -2 & -1 & -7 & 0 \end{array} \right) \xrightarrow{R_1 + R_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 6 & 0 \\ -1 & -2 & -1 & -7 & 0 \end{array} \right) \\ & \xrightarrow{R_1 + R_3} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 6 & 0 \\ 0 & 0 & -2 & -6 & 0 \end{array} \right) \xrightarrow{R_1 + R_3} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 2 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \\ & \xrightarrow{\frac{1}{2} R_2} \left(\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_2 + R_1} \left(\begin{array}{cccc|c} 1 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

Put $w = t$ (t , is real number), then,

$$\text{From the 2-nd row we get, } z + 3w = 0 \Rightarrow z = -3w \Rightarrow z = -3t.$$

Put $y = s$ (s , is real number), then

$$\text{From the 1-st row we get, } x + 2y + 4w = 0 \Rightarrow x = -2y - 4w \Rightarrow x = -2s - 4t.$$

Therefore, the solution set is : $(x = -2s - 4t, y = s, z = -3t, w = t)$

Example (4): Solve the following system of equations.

$$x + y + z + w = 0$$

$$x + w = 0$$

$$x + 2y + z = 0$$

Solution

Here, in this example we notice that the number of variables (unknowns) is greater than the number of equations, then the system has an infinite number of solutions

Therefore,

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{array} \right) \xrightarrow{-R_1 + R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{array} \right) \xrightarrow{-R_1 + R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \end{array} \right) \xrightarrow{R_2 + R_3}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \end{array} \right) \xrightarrow{R_3 + R_1} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \end{array} \right) \xrightarrow{-R_2} \left(\begin{array}{cccc|c} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & -1 & 0 \end{array} \right) \xrightarrow{-R_3}$$

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array} \right), \text{ then, the system is consistent and has an infinite number of solutions.}$$

Put $w = r$ (r is real number), then,

$$\text{From the 3-rd row we get, } z + w = 0 \Rightarrow z = -w \Rightarrow z = -r.$$

$$\text{From the 2-nd row we get, } y + z = 0 \Rightarrow y = -z \Rightarrow y = -(-r) = r.$$

$$\text{From the 1-st row we get, } x + y = 0 \Rightarrow x = -y \Rightarrow x = -r.$$

Therefore, the solution set is : $(x = -r, y = r, z = -r, w = r)$